



**COMPETENCY STANDARD
FOR
PRODUCTION PROCESS IN WET PROCESSING
(TEXTILE SECTOR)**

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance
Government of the People's Republic of Bangladesh

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The Competency Standard for Production Process in Wet Processing is designed to guide the development of curricula, teaching materials, and assessment tools. It also serves as the document for providing training consistent with the requirements of the industry, so that individuals who pass through the set standard via assessment would be qualified and settled for a relevant job.

This document has been developed to be used in training under the Skills for Industry Competitiveness and Innovation Program (SICIP) to meet the skills requirements of the Textile sector. This document is owned by the Finance Division of the Ministry of Finance of the People's Republic of Bangladesh.

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INTRODUCTION:

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades, which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

The textile sector is the largest manufacturing and export-earning sector of Bangladesh, and wet processing – covering pre-treatment, dyeing, printing and finishing – is the stage of the textile value chain where colour, appearance, hand feel and performance are imparted to the material. It is also the most resource-intensive stage of the value chain, consuming the largest share of water, chemicals and thermal energy and generating the greater part of the industry's effluent load. The quality, cost competitiveness and environmental compliance of the whole downstream garment industry therefore depend largely on the skills of the workforce engaged in the wet processing production process. This Competency Standard has accordingly been developed to prepare a competent, safety-conscious and environmentally responsible entry-level workforce for the production process in wet processing.

The Competency Standard also suggests integration of YouTube or similar platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and to ensure that the required skills are aligned with industry requirements.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competency and its corresponding Elements.

OVERVIEW:

A Competency Standard is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

1. provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
2. enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies.
3. encourages the development and delivery of flexible training that suits individual and industry requirements.
4. encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standards are developed by a working group comprised of national process experts and the participation of experts from the industry to identify the competencies required of an occupation in a particular sector.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at a training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

1. Reference to Industry Sector, Occupational Title and Occupational Description
2. Unit code
3. Unit title
4. Unit descriptor
5. Unit of Competency
6. Elements and performance criteria
7. Variables and range statement
8. Evidence guides

Together all the parts of a Unit of Competency:

1. Describe a work activity
2. Guide the assessor in determining whether the candidate is competent.

This Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution and relevant consultants of SICIP.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competency:

1. An overview of all Units of Competency for the occupation and their corresponding duration required for completion of training.
2. The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

DESCRIPTION OF THE OCCUPATION AND THE TRAINING COURSE:

Industry Sector	Textile
Occupation Title	Production Process in Wet Processing
Occupational Description	A worker competent in the Production Process in Wet Processing carries out the sequential operations required to convert grey yarn, knit or woven fabric into dyed, printed and finished textile material in a wet processing (dyeing–printing–finishing) unit. The occupation covers the handling, weighing and preparation of dyes, chemicals and auxiliaries; pre-treatment operations such as singeing, desizing, scouring, bleaching and mercerizing; dyeing of yarn and fabric; printing; mechanical and chemical finishing including stentering and heat setting; in-process quality checking; routine machine care and housekeeping; and support to effluent handling and environmental compliance. The worker performs these tasks under supervision, following standard operating procedures, batch cards, recipes, safety data sheets, safety requirements and the environmental practices of the mill, and records production and quality data using workplace formats.
Level of the Course	Basic Level Skills Training
Nominal Duration	360 Hours (03 Months)
Entry Qualification of Trainees	Grade VIII passed; physically and mentally fit to work in a wet processing environment involving heat, steam, moisture, chemicals and wet floors; minimum age as per the prevailing labour law of Bangladesh.
Target Number of Trainees	A total of 2,000 trainees to be trained by the end of 2028
Training Providers	Textile industries and training institutes having the capacity to deliver and run the training program in accordance with this Competency Standard and the Workshop/Lab Facility Standard.
Mode of Delivery	Institution-based and/or industry-based competency-based training combining theory, demonstration, guided practice and practical performance in an actual or simulated workplace.
Employment / Job Placement	The training providers (industries and training institutes) will arrange employment for at least 65% of the total enrollees upon completion of the training.
Assessment and Certification	Competency assessment will be conducted as per the Assessment Evidence Guide of each unit, in an assessment/training centre or in an actual or simulated workplace, by a nationally certified assessor or occupation-specific industry expert.
Possible Job Titles	Batch Preparation Helper, Pre-treatment Machine Operator/Helper, Dyeing Machine Operator (Winch/Jigger/Jet/Soft-flow), Yarn Dyeing Operator, Colour Kitchen Assistant, Sample/Lab Dip Assistant, Print Paste Kitchen Assistant, Printing Machine/Table Helper, Stenter Operator/Helper, Finishing Machine Operator, Fabric Inspector/Quality Checker (Trainee level), ETP Operator Assistant.

UNITS & ELEMENTS AT A GLANCE:

Generic Competencies (20 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-TEX-PPWP-01-G	Apply Occupational Health and Safety (OHS) Practices in the Workplace	1. Identify OHS policies and procedures 2. Apply chemical safety and environmental compliance practices in wet processing 3. Conserve water, steam and energy in wet processing 4. Support effluent handling and treatment	10
SICIP-TEX-PPWP-02-G	Work in a Team Environment	1. Identify team goals and processes 2. Communicate and cooperate with team members 3. Work as a team member 4. Solve problems as a team member	10
Total Hours			20 Hrs.

Sector Specific Competencies (20 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-TEX-PPWP-01-S	Work in the Textile Sector	1. Identify the textile sector and its main activities 2. Understand basic roles in textile production processes 3. Recognize common tools and equipment used 4. Understand simple quality requirements in textile work	10
SICIP-TEX-PPWP-02-S	Apply Green Practices in the Textile Sector	1. Interpret green concepts 2. Minimize resource use in the workplace 3. Implement waste management practices	10
Total Hours			20 Hrs.

Occupation Specific Competencies (320 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-TEX-PPWP-01-O	Understand Fundamentals of Textile Wet Processing	1. Interpret textile wet processing and materials to be processed 2. Identify dyes, chemicals and auxiliaries used in wet processing 3. Identify machines, tools and equipment used in wet processing 4. Describe the wet processing sequence 5. Interpret work instructions, recipes and production documents	30
SICIP-TEX-PPWP-02-O	Apply Dyes, Chemicals and Auxiliaries for Wet Processing	1. Prepare for chemical handling work 2. Store and handle dyes, chemicals and auxiliaries safely 3. Weigh, dissolve and prepare dyes, chemicals and auxiliaries 4. Dispense and dose solutions to the machine	20
SICIP-TEX-PPWP-03-O	Perform Yarn and Fabric Pre-treatments	1. Prepare grey material and machines for pre-treatment 2. Carry out singeing and de-sizing operations 3. Carry out scouring, bleaching and bio-polishing operations 4. Carry out mercerizing operations 5. Check pre-treatment quality and record process data	65
SICIP-TEX-PPWP-04-O	Perform Dyeing Operations	1. Prepare the batch and dyeing machine 2. Prepare and charge the dye bath 3. Carry out the dyeing operation 4. Perform after-treatment, washing-off and soaping	90

		5. Check shade and dyeing quality and record data	
SICIP-TEX-PPWP-05-O	Perform Printing Operations	1. Prepare for printing operations 2. Prepare print paste 3. Carry out printing operations 4. Carry out drying, fixation and washing-off after printing 5. Check print quality and record data	50
SICIP-TEX-PPWP-06-O	Perform Finishing Operations	1. Prepare the fabric and finishing machines for finishing 2. Carry out mechanical finishing operations 3. Carry out chemical finishing operations 4. Perform stentering, heat setting and dimensional stabilization 5. Check finished fabric quality and prepare for delivery	50
SICIP-TEX-PPWP-07-O	Perform In-process Quality Checks and Routine Machine Care	1. Perform in-process quality checks and basic tests 2. Record and report production and quality data 3. Maintain housekeeping and workplace conditions	15
Total Hours			320 Hrs.

Grand Total Duration of the Training Course	360 Hours (03 Months)
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COMPETENCY STANDARD: PRODUCTION PROCESS IN WET PROCESSING

The Generic Competencies

Unit of Competency: APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICES IN THE WORKPLACE	Nominal Duration: 10 Hrs.	Unit Code: SICIP-TEX-PPWP-01-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply Occupational Health and Safety (OHS) practices in the workplace. It specifically includes the tasks of identifying OHS policies and procedures, applying chemical safety and environmental compliance practices in wet processing, conserving water, steam and energy in wet processing, and supporting effluent handling and treatment.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify OHS policies and procedures	1. 1 <u>OHS policies</u> and safe operating procedures are read and understood. 1. 2 Safety signs and symbols are identified and followed. 1. 3 Emergency response, evacuation procedures and other contingency measures are determined.
2. Apply chemical safety and environmental compliance practices in wet processing	2. 1 <u>Chemical exposure control measures</u> are applied during handling and processing. 2. 2 Chemical emergency is responded to following the workplace emergency procedure. 2. 3 <u>Compliance standards and certifications</u> applicable to wet processing are recognized. 2. 4 Records supporting chemical compliance and traceability are maintained.
3. Conserve water, steam and energy in wet processing	3. 1 Water consumption is monitored and controlled during processing. 3. 2 <u>Steam and thermal energy saving practices</u> are applied wherever possible. 3. 3 Leakage of water, steam, compressed air and chemicals is identified and reported promptly. 3. 4 <u>Heat and water recovery systems</u> are operated or supported as per the level of authority.
4. Support effluent handling and treatment	4.1 <u>Sources and characteristics of wet processing effluent</u> are identified. 4.2 Effluent is directed to the correct collection and segregation system.

	4.3 <u>Basic ETP processes and units</u> are identified and their functions are described. 4.4 <u>Effluent discharge parameters</u> are recognized and compared with the standard.
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Range of Variables:

Variable	Range (May include but not limited to)
1. OHS policies	1. 1 International OHS requirements 1. 2 Bangladesh standards for OHS 1. 3 Bangladesh Labour Act and Rules 1. 4 Building Code 1. 5 Fire Safety Rules and Regulations 1. 6 Bangladesh Environment Conservation Act and Rules 1. 7 Industry guidelines and mill safety manual 1. 8 Chemical handling and storage policy of the organization
2. Chemical exposure control measures	2. 1 Engineering controls – closed dosing systems, local exhaust ventilation, bunding and containment 2. 2 Administrative controls – SOPs, job rotation, training, signage and restricted access 2. 3 Personal protective equipment – goggles, chemical gloves, respirator, apron and boots 2. 4 Prohibition of eating, drinking and smoking in chemical areas 2. 5 Handwashing, showering and decontamination facilities 2. 6 Ventilation of the chemical store, colour kitchen and print paste kitchen
3. Compliance standards and certifications	3. 1 ZDHC (Zero Discharge of Hazardous Chemicals) MRSL and wastewater guidelines 3. 2 OEKO-TEX Standard 100 3. 3 GOTS (Global Organic Textile Standard) 3. 4 REACH regulation of the European Union 3. 5 Buyer-specific restricted substance list (RSL) 3. 6 Bangladesh Environment Conservation Act and Environment Conservation Rules 3. 7 Environmental clearance certificate from the Department of Environment (DoE) 3. 8 ISO 14001 environmental management system 3. 9 ISO 45001 occupational health and safety management system
4. Steam and thermal energy saving practices	4.1 Insulation of steam pipes, valves, flanges and vessels 4.2 Condensate recovery and return to the boiler feed tank 4.3 Steam trap maintenance and prompt repair of steam leaks 4.4 Optimum heating and cooling gradients and avoidance of over-heating

	4.5 Use of hot water recovered through heat exchangers for the next process 4.6 Keeping machine lids and doors closed during heating 4.7 Scheduling of batches to reduce repeated machine heating cycles 4.8 Optimum stenter zone temperature, speed and exhaust damper setting 4.9 Boiler efficiency, blowdown control and flue gas monitoring
5. Heat and water recovery systems	5.1 Heat exchanger for recovering heat from hot effluent 5.2 Condensate recovery system and flash steam recovery 5.3 Hot water storage tank and reuse arrangement 5.4 Economizer and air pre-heater on the boiler 5.5 Stenter exhaust air heat recovery unit 5.6 Caustic recovery plant for the mercerizing process 5.7 Reverse osmosis (RO) plant and treated water recycling 5.8 Cooling tower and closed-loop cooling water circuit
6. Sources and characteristics of wet processing effluent	6.1 Desizing effluent – high BOD and COD from starch and size 6.2 Scouring effluent – strongly alkaline, high in wax, pectin, oil and impurities 6.3 Bleaching effluent – residual peroxide, hypochlorite and stabilizers 6.4 Mercerizing effluent – strongly alkaline with high caustic content 6.5 Dyeing effluent – strong colour, salt, unfixed dye and high TDS 6.6 Printing effluent – thickener, urea, pigment, binder and solvent 6.7 Finishing effluent – softener, resin, formaldehyde and repellent chemicals 6.8 Washing and soaping effluent – large volume with moderate load 6.9 Effluent characteristics – pH, colour, temperature, BOD, COD, TDS, TSS, oil and grease, heavy metals and sulphide
7. Basic ETP processes and units	7. 1 Equalization tank with mixing arrangement 7. 2 Neutralization and pH correction unit 7. 3 Cooling tower or cooling arrangement for hot effluent 7. 4 Coagulation and flocculation with flash mixer and chemical dosing 7. 5 Primary clarifier and settling tank 7. 6 Biological treatment – aeration tank, activated sludge, MBBR or SBR 7. 7 Secondary clarifier

	7. 8 Tertiary treatment – sand filter, activated carbon filter and ozonation 7. 9 Reverse osmosis and water recovery unit where installed 7. 10 Sludge thickener, drying bed and filter press 7. 11 Sludge storage and authorized disposal arrangement 7. 12 Effluent flow meter and online monitoring system
8. Effluent discharge parameters	8.1 pH – Bangladesh inland surface water discharge standard, generally 6–9 8.2 BOD₅ – generally not exceeding 50 mg/l 8.3 COD – generally not exceeding 200 mg/l 8.4 Total suspended solids (TSS) – generally not exceeding 150 mg/l 8.5 Total dissolved solids (TDS) 8.6 Colour measured in Pt-Co (Hazen) units 8.7 Temperature of the discharged effluent 8.8 Oil and grease content 8.9 Heavy metals – chromium, copper, zinc, lead and nickel 8.10 Sulphide and residual chlorine 8.11 Standards prescribed under the Bangladesh Environment Conservation Rules and monitored by the Department of Environment (DoE)

Curricular Content Guide:

1. Underpinning Knowledge	1. 1 OHS workplace policies and procedures. 1. 2 Work safety procedures in a textile wet processing unit. 1. 3 Chemical emergency response, first aid for chemical injuries and incident reporting. 1. 4 Water consumption benchmarks in wet processing and the factors that influence them. 1. 5 Steam generation, distribution, condensate recovery and steam saving techniques. 1. 6 Thermal energy consumption of the stenter, dryers and dyeing machines. 1. 7 Heat and water recovery systems and caustic recovery. 1. 8 Sources, volume and characteristics of effluent from each wet processing stage. 1. 9 Effluent segregation, collection and pre-treatment requirements. 1. 10 Effluent treatment plant – units, processes and their functions. 1. 11 Physical, chemical and biological treatment principles. 1. 12 International compliance standards and certification schemes applicable to wet processing.
2. Underpinning Skills	2. 1 Identifying OHS policies and procedures 2. 2 Following personal work safety practices 2. 3 Selecting and using appropriate PPE for chemical, thermal and wet work

	2. 4 Responding to chemical splashes, spills and gas releases 2. 5 Reading and recording water, steam and electricity meters 2. 6 Identifying and reporting leakage of water, steam, air and chemicals 2. 7 Supporting the operation of heat and water recovery systems 2. 8 Identifying the effluent generated by each process and directing it correctly 2. 9 Segregating concentrated liquor and hazardous effluent 2. 10 Identifying ETP units and describing their functions 2. 11 Checking chemicals against the approved chemical list before use 2. 12 Maintaining chemical compliance and traceability records
3. Underpinning Attitudes	3. 1 Commitment to occupational health and safety 3. 2 Promptness in carrying out activities 3. 3 Sincere and honest to duties 3. 4 Environmental concerns 3. 5 Eagerness to learn 3. 6 Tidiness and timeliness 3. 7 Respect for rights of peers and seniors in the workplace 3. 8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 PPEs (chemical and thermal) 4.3 Firefighting equipment 4.4 First aid box and emergency response manual 4.5 Water, steam and electricity meters with reading records 4.6 Access to an effluent treatment plant or a model/demonstration unit 4.7 pH meter, thermometer, jar test apparatus and sampling bottles 4.8 Effluent testing kit or laboratory access for BOD, COD and TSS 4.9 Colour-coded waste bins and hazardous waste storage arrangement 4.10 Bangladesh Environment Conservation Rules discharge standard charts 4.11 SOPs, environmental policy and compliance documents

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1. 1 Identified and followed OHS policies and procedures. 1. 2 Selected and used personal protective equipment (PPE).
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	1. 3 Applied chemical safety and environmental compliance practices in wet processing 1. 4 Conserved water, steam and energy in wet processing 1. 5 Supported effluent handling and treatment
2. Methods of Assessment	Competency should be assessed by: 2. 1 Written test 2. 2 Practical demonstration 2. 3 Oral questioning 2. 4 Portfolio (Optional)
3. Context of Assessment	3. 1 Competency assessment must be done in an assessment/training centre or in an actual or simulated workplace after completion of the training. 3. 2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: WORK IN A TEAM ENVIRONMENT	Nominal Duration: 10 Hrs.	Unit Code: SICIP-TEX-PPWP-02-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to work in a team environment. It specifically includes the tasks of identifying team goals and processes, communicating and cooperating with team members, working as a team member and solving problems as a team member.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify team goals and processes	1.1 Team goals and collaborative decision-making processes are identified. 1.2 Roles and responsibilities of team members are identified. 1.3 Relationships within the team and with other workers are identified.
2. Communicate and cooperate with team members	2. 1 Effective interpersonal skills are used to interact with team members and to contribute to activities and objectives. 2. 2 Formal and informal <u>forms of communication</u> are used effectively to support team achievement. 2. 3 Diversity in character is respected and valued in team functioning. 2. 4 Views and opinions of other team members are understood and valued. 2. 5 Workplace terminology is used correctly to assist communication.
3. Work as a team member	3.1 Duties, responsibilities, authorities, objectives and task requirements are identified and clarified with the team. 3.2 Tasks are performed in accordance with organizational and team requirements, specifications, and workplace procedures. 3.3 Support to other team members is provided to ensure the team achieves its goals. 3.4 Agreed reporting lines are followed using standard operating procedures.
4. Solve problems as a team member	4.1 Current and potential problems faced by the team are identified. 4.2 A solution to the problem is identified. 4.3 Problems are solved effectively and the outcome of the implemented solution is evaluated.

Range of Variables:

Variable	Range (May include but not limited to)
1. Forms of communication	1. 1 Agenda 1. 2 Simple reports such as progress and incident reports 1. 3 Job sheets and shift handover log 1. 4 Operational manuals and SOPs 1. 5 Brochures and promotional material 1. 6 Visual and graphic materials 1. 7 Standards 1. 8 OSH information 1. 9 Signs and symbols

Curricular Content Guide:

1. Underpinning Knowledge	1. 1 Team goals and collaborative decision-making processes 1. 2 Roles and responsibilities of team members 1. 3 Relationships within the team and with other workers 1. 4 Effective interpersonal skills to interact with team members 1. 5 Effective formal and informal forms of communication 1. 6 Value of diversity in team functioning 1. 7 Correct use of workplace terminology 1. 8 Team's duties, responsibilities, authorities, objectives and task requirements 1. 9 Support mechanisms for other members of the team to ensure achievement of goals 1. 10 Methods of identifying current and potential problems faced by a team 1. 11 Effective problem-solving methods and evaluation of outcomes
2. Underpinning Skills	2. 1 Identifying team goals and collaborative decision-making processes 2. 2 Identifying roles and responsibilities of team members 2. 3 Using effective interpersonal skills to interact with team members 2. 4 Using formal and informal forms of communication 2. 5 Understanding and valuing views and opinions of other team members 2. 6 Performing tasks in accordance with organizational and team requirements, specifications and workplace procedures 2. 7 Supporting other members of the team to ensure the team achieves its goals 2. 8 Identifying current and potential problems faced by the team 2. 9 Identifying solutions to problems

	2. 10 Solving problems effectively and evaluating the outcome of the implemented solution
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Manuals/Guidelines 4.3 Pens 4.4 Papers 4.5 Workbooks

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1. 1 Identified team goals and work processes. 1. 2 Communicated and cooperated with team members. 1. 3 Worked as a team member. 1. 4 Solved problems as a team member.
2. Methods of Assessment	Competency should be assessed by: 2. 1 Written test 2. 2 Practical demonstration 2. 3 Oral questioning 2. 4 4. Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Sector Specific Competencies

Unit of Competency: WORK IN THE TEXTILE SECTOR	Nominal Duration: 10 Hrs.	Unit Code: SICIP-TEX-PPWP-01-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required for work in the textile sector. It specifically includes the tasks of identifying the textile sector and its main activities, understanding basic roles in textile production processes, recognizing common tools and equipment used, and understanding simple quality requirements in textile work.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify the textile sector and its main activities	1. 1 The textile sector is identified within the context of national and global industries. 1. 2 <u>Main activities</u> of the textile sector are recognized. 1. 3 <u>Different stages of the textile value chain</u> are described in simple terms. 1. 4 Key workplaces in the textile sector are identified. 1. 5 Importance of the textile sector in the economy and the environment is understood.
2. Understand basic roles in textile production processes	2. 1 <u>Basic job roles in textile production</u> are identified and described. 2. 2 Simple production workflows and task sequences are understood. 2. 3 Teamwork and coordination in production processes are identified. 2. 4 Workplace safety roles and responsibilities are understood and followed. 2. 5 Shift working practices and reporting lines are understood.
3. Recognize common tools and equipment used	3.1 <u>Common tools used in textile processing</u> are identified correctly. 3.2 <u>Basic machines and equipment used in production</u> are recognized. 3.3 Functions of tools and equipment are understood in simple terms. 3.4 Basic maintenance and cleaning needs of tools and equipment are recognized.
4. Understand simple quality requirements in textile work	4.1 Basic quality standards in textile work are identified. 4.2 <u>Simple defects in textile products</u> are recognized. 4.3 Importance of maintaining quality during production is understood.

Elements of Competency	Performance Criteria
	4.4 Work output is checked against basic quality expectations. 4.5 Good workmanship practices are followed to ensure quality.

Range of Variables:

Variable	Range (May include but not limited to)
1. Main activities	1. 1 Fibre production and preparation 1. 2 Yarn manufacturing (spinning) 1. 3 Fabric manufacturing (weaving and knitting) 1. 4 Wet processing (dyeing, printing and finishing) 1. 5 Garment manufacturing 1. 6 Testing and quality assurance 1. 7 Packaging, storage and export
2. Different stages of the textile value chain	2. 1 Natural and man-made fibre production 2. 2 Fibre preparation and mixing 2. 3 Yarn manufacturing (spinning) 2. 4 Fabric manufacturing (weaving/knitting) 2. 5 Dyeing, printing and finishing 2. 6 Cutting, sewing and finishing of garments 2. 7 Packaging, distribution and export
3. Basic job roles in textile production	3.1 Blow room operator/helper 3.2 Card tenter 3.3 Draw frame and comber operator 3.4 Simplex (speed frame) operator 3.5 Ring frame sider and doffer 3.6 Winding and packing operator 3.7 Quality checker/laboratory assistant 3.8 Maintenance fitter/electrician 3.9 Line leader/supervisor
4. Common tools used in textile processing	4.1 Sliver and roving cutter, scissors and nippers 4.2 Hooks, brushes and dusting tools 4.3 Spanner, screwdriver and plier set 4.4 Oil can and grease gun 4.5 Feeler and setting gauges 4.6 Wrap reel, wrap block and electronic balance 4.7 Tachometer and hand lens 4.8 Vacuum cleaner and air blowing gun
5. Basic machines and equipment used in production	5. 1 Bale opener/plucker and blow room line 5. 2 Carding machine 5. 3 Draw frame (breaker and finisher) 5. 4 Lap former and comber 5. 5 Speed frame (simplex)

Variable	Range (May include but not limited to)
	5. 6 Ring frame 5. 7 Winding machine/autoconer 5. 8 Humidification and waste collection plant 5. 9 Testing laboratory equipment
6. Simple defects in textile products	6. 1 Neps and slubs 6. 2 Thick and thin places 6. 3 Count variation and uneven yarn 6. 4 High hairiness 6. 5 Foreign fibre and contamination 6. 6 Sliver and roving breakage 6. 7 Soft/hard and damaged packages 6. 8 Oil and dirt stains

Curricular Content Guide:

1. Underpinning Knowledge	1. 1 Structure and importance of the textile sector in the national and global economy. 1. 2 Main activities and stages of the textile value chain. 1. 3 Position of yarn manufacturing within the textile value chain. 1. 4 Basic job roles, responsibilities and reporting lines in textile production. 1. 5 Simple production workflows, task sequences and shift practices. 1. 6 Common tools, machines and equipment used in textile production and their functions. 1. 7 Basic maintenance and cleaning needs of tools and equipment. 1. 8 Basic quality standards and common defects in textile products. 1. 9 Importance of quality, productivity and good workmanship in the workplace.
2. Underpinning Skills	2. 1 Identifying the textile sector and its main activities 2. 2 Describing the stages of the textile value chain in simple terms 2. 3 Understanding basic job roles and responsibilities in textile production processes 2. 4 Following simple production workflows, task sequences and teamwork practices 2. 5 Applying workplace safety roles and responsibilities in production activities 2. 6 Recognizing common tools, machines and equipment and their functions 2. 7 Recognizing basic maintenance and cleaning needs of tools and equipment

	2. 8 Recognizing simple defects and checking work output against basic quality requirements 2. 9 Following good workmanship practices to ensure consistent quality output
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Charts, models and samples of fibre, sliver, roving and yarn 4.3 Multimedia projector, laptop and audio-visual clips 4.4 Learning manual and handouts 4.5 Stationery

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1. 1 Identified the textile sector and its main activities. 1. 2 Understood basic roles in textile production processes. 1. 3 Recognized common tools and equipment used. 1. 4 Understood simple quality requirements in textile work.
2. Methods of Assessment	Competency should be assessed by: 2. 1 Written test 2. 2 Practical demonstration 2. 3 Oral questioning 2. 4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: APPLY GREEN PRACTICES IN THE TEXTILE SECTOR	Nominal Duration: 10 Hrs.	Unit Code: SICIP-TEX-PPWP-02-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply green practices in the textile sector. It specifically includes the tasks of interpreting green concepts, minimizing resource use in the workplace and implementing waste management practices.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret green concepts	1. 1 Basic concepts of environmental sustainability and green practices are explained clearly. 1. 2 <u>Environmental impacts of textile production</u> processes are identified. 1. 3 <u>Resource efficiency concepts</u> are interpreted. 1. 4 Waste reduction, reuse and recycling principles are described in relation to yarn manufacturing operations. 1. 5 <u>Eco-friendly practices in the textile sector</u> are recognized and related to workplace activities.
2. Minimize resource use in the workplace	2. 1 Electricity, water, compressed air and raw materials are used efficiently according to workplace requirements. 2. 2 <u>Unnecessary consumption of resources</u> is identified and avoided during operations. 2. 3 Machines and equipment are operated in energy-efficient modes where applicable. 2. 4 Material wastage is minimized through careful handling and accurate measurement. 2. 5 <u>Resource-saving practices</u> are applied where possible. 2. 6 Leakage, spillage and wastage of materials are promptly controlled and reported. 2. 7 <u>Workplace practices</u> for resource conservation are followed.
3. Implement waste management practices	3.1 <u>Types of waste</u> generated in yarn manufacturing operations are identified correctly. 3.2 Waste materials are segregated according to workplace and environmental requirements. 3.3 <u>Reusable and recyclable materials</u> are separated from non-recyclable waste. 3.4 Waste is stored safely in designated containers or areas. 3.5 Waste disposal is carried out following environmental rules and organizational procedures.

Elements of Competency	Performance Criteria
	3.6 Hazardous waste is handled carefully according to safety guidelines. 3.7 Waste reduction practices are applied to minimize environmental impact during operations.

Range of Variables:

Variable	Range (May include but not limited to)
1. Environmental impacts of textile production	1. 1 High electrical energy consumption of spinning machinery and humidification plants 1. 2 Cotton dust, fibre fly and micro-fibre emission 1. 3 Noise pollution from machines and compressors 1. 4 Solid waste generation (droppings, comber noil, soft and hard waste) 1. 5 Water consumption and effluent from wet processing 1. 6 Air emission and greenhouse gases from boilers and generators 1. 7 Used oil, grease and contaminated cotton waste
2. Resource efficiency concepts	2. 1 Energy-efficient motors and inverter/variable frequency drives 2. 2 Control of compressed air leakage and optimized air pressure 2. 3 Optimized humidification and ventilation control 2. 4 Reduction of soft and hard waste percentage 2. 5 Preventive maintenance to reduce power consumption 2. 6 Energy-efficient lighting and switching-off practices 2. 7 Optimum machine loading and reduction of idle running
3. Eco-friendly practices in the textile sector	3.1 Recycling of soft waste back into mixing where permitted 3.2 Use of recycled, organic or sustainably sourced fibre 3.3 Dust collection, filter and waste recovery systems 3.4 Compliance with Department of Environment (DoE) rules and regulations 3.5 Effluent treatment plant (ETP) and rainwater harvesting 3.6 Reuse of cans, bobbins, cones, cartons and packaging materials 3.7 Green building and energy audit practices
4. Unnecessary consumption of resources	4.1 Idle running of machines and motors 4.2 Compressed air leakage in pipelines and guns 4.3 Excess lighting in unoccupied areas 4.4 Over-cleaning with air guns instead of vacuum 4.5 Excess lubricant and cleaning material use 4.6 Excessive good fibre loss with droppings and noil

Variable	Range (May include but not limited to)
5. Resource-saving practices	5. 1 Switching off machines, lights and fans when not required 5. 2 Reporting air, water and oil leakage promptly 5. 3 Using vacuum cleaning instead of air blowing where possible 5. 4 Reusing cans, tubes, cones and packaging materials 5. 5 Maintaining correct machine settings to reduce waste 5. 6 Recording and monitoring waste and energy use
6. Workplace practices	6. 1 Standard operating procedures (SOPs) 6. 2 Environmental policy of the organization 6. 3 Housekeeping and 5S practices 6. 4 Energy and waste monitoring records 6. 5 Instructions from supervisors and safety notices
7. Types of waste	7.1 Blow room droppings and card droppings/flat strips 7.2 Comber noil 7.3 Sliver, roving and pneumafil soft waste 7.4 Hard waste from winding (yarn and thread waste) 7.5 Damaged cans, bobbins, cones and tubes 7.6 Packaging waste (poly bags, bale wrappers, straps, cartons) 7.7 Used oil, grease and oil-contaminated cotton (hazardous waste) 7.8 Used filters, sweeps and floor waste
8. Reusable and recyclable materials	8.1 Usable soft waste for back-mixing 8.2 Comber noil for coarse count and open-end spinning 8.3 Cans, bobbins, tubes and cones 8.4 Cartons, poly bags and bale straps 8.5 Metal and plastic scrap 8.6 Paper and stationery waste

Curricular Content Guide:

1. Underpinning Knowledge	1. 1 Basic concepts of environmental sustainability, green practices and climate-resilient production. 1. 2 Environmental impacts of textile and yarn manufacturing operations. 1. 3 Resource efficiency concepts relating to energy, water, compressed air and raw material. 1. 4 Principles of waste reduction, reuse and recycling in a spinning mill. 1. 5 Eco-friendly practices and technologies applied in the textile sector. 1. 6 Types of waste generated at each stage of yarn manufacturing and permissible waste levels.
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	1. 7 Waste segregation, storage, disposal methods and compliance with environmental regulations. 1. 8 Hazardous waste handling and workplace environmental safety procedures. 1. 9 Relevant national environmental rules and organizational environmental policy.
2. Underpinning Skills	2. 1 Interpreting green concepts and relating them to workplace activities 2. 2 Identifying environmental impacts of yarn manufacturing operations 2. 3 Applying resource efficiency principles in workplace operations 2. 4 Using electricity, compressed air, water and raw materials efficiently 2. 5 Identifying and avoiding unnecessary consumption of resources 2. 6 Operating machines and equipment in energy-efficient modes where applicable 2. 7 Handling materials carefully to reduce wastage and improve accuracy 2. 8 Controlling and reporting leakage, spillage and resource wastage promptly 2. 9 Identifying and segregating different types of waste generated in yarn manufacturing 2. 10 Separating reusable and recyclable materials from non-recyclable waste 2. 11 Handling hazardous waste according to safety guidelines and regulations
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4. 1 Workplace (simulated or actual) 4. 2 Standard operating procedures 4. 3 Workplace documents, signs and symbols 4. 4 Codes of conduct and environmental policy 4. 5 Waste segregation bins and storage area 4. 6 Tools, equipment and facilities appropriate to the process or activities 4. 7 Materials relevant to the proposed activity

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1. 1 Interpreted green concepts. 1. 2 Minimized resource use in the workplace. 1. 3 Implemented waste management practices.
2. Methods of Assessment	Competency should be assessed by: 2. 1 Written test 2. 2 Practical demonstration 2. 3 Oral questioning 2. 4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Occupation Specific Competencies

Unit of Competency: UNDERSTAND FUNDAMENTALS OF TEXTILE WET PROCESSING	Nominal Duration: 30 Hrs.	Unit Code: SICIP-TEX-PPWP-01-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to understand fundamentals of textile wet processing. It specifically includes the tasks of interpreting textile wet processing and materials to be processed, identifying dyes, chemicals and auxiliaries used in wet processing, identifying machines, tools and equipment used in wet processing, describing the wet processing sequence, and interpreting work instructions, recipes and production documents.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret textile wet processing and materials to be processed	1. 1 General concept of textile wet processing is understood. 1. 2 <u>Types of textile fibres</u> are identified based on source and characteristics. 1. 3 <u>Yarn and fabric structures</u> presented for wet processing are identified. 1. 4 Physical and chemical properties of fibres relevant to wet processing are described. 1. 5 Grey material faults affecting wet processing are recognized.
2. Identify dyes, chemicals and auxiliaries used in wet processing	2. 1 <u>Classes of dyes</u> are identified and related to the fibres they are applied on. 2. 2 <u>Basic chemicals</u> used in wet processing are identified with their functions. 2. 3 <u>Auxiliaries</u> are identified with their functions. 2. 4 <u>Pigments and printing chemicals</u> are identified. 2. 5 Safety data sheet (SDS/MSDS) information and hazard pictograms are interpreted.
3. Identify machines, tools and equipment used in wet processing	3. 1 <u>Machines used in wet processing</u> are identified. 3. 2 Machines are classified according to process stage and material form. 3. 3 <u>Laboratory and measuring instruments</u> are identified and their functions are described. 3. 4 Utilities and services required for wet processing are identified. 3. 5 Safety requirements for using machines, tools and equipment are identified.

4. Describe the wet processing sequence	4.1 <u>Stages of the wet processing sequence</u> are described in the correct order. 4.2 Objectives of each process stage are explained. 4.3 Sequence variations for knit, woven and yarn processing are explained. 4.4 <u>Process parameters</u> influencing quality are described. 4.5 Water quality requirements for wet processing are described.
5. Interpret work instructions, recipes and production documents	5.1 Production documents - recipes, batch cards and process cards are read and understood. 5.2 <u>Basic process calculations</u> are performed correctly. 5.3 Standard operating procedures (SOPs) are followed as instructed. 5.4 Information is recorded and reported using workplace formats. 5.5 Workplace terminology, units and symbols are used correctly.

Range of Variables:

Variable	Range (May include but not limited to)
1. Types of textile fibres	1.1 Natural fibres 1.1.1 Cotton 1.1.2 Jute 1.1.3 Wool 1.1.4 Silk 1.1.5 Linen 1.2 Man-made fibres 1.2.1 Polyester 1.2.2 Nylon (polyamide) 1.2.3 Acrylic 1.2.4 Polypropylene 1.3 Regenerated fibres 1.3.1 Viscose rayon 1.3.2 Modal 1.3.3 Lyocell/Tencel 1.4 Elastomeric fibre (spandex/elastane) 1.5 Blends 1.5.1 Polyester-cotton (PC) 1.5.2 Chief value cotton (CVC) 1.5.3 Poly-viscose (PV) 1.5.4 Cotton-elastane and polyester-elastane 1.6 Recycled and organic fibres
2. Yarn and fabric structures	2.1 Yarn in hank, cone, cheese and beam form 2.2 Single jersey, interlock, rib, pique, fleece and lycra knit fabrics 2.3 Plain, twill, satin, dobby and jacquard woven fabrics

	2. 4 Denim, terry towel and corduroy fabrics 2. 5 Non-woven fabrics 2. 6 Fabric parameters – GSM, width, construction, ends and picks, wales and courses 2. 7 Tubular and open-width fabric forms
3. Classes of dyes	3. 1 Reactive dyes (cold brand, hot brand, bi-functional) 3. 2 Direct dyes 3. 3 Vat dyes 3. 4 Sulphur dyes 3. 5 Indigo (for denim) 3. 6 Azoic/naphthol dyes 3. 7 Disperse dyes 3. 8 Acid and metal complex dyes 3. 9 Basic (cationic) dyes 3. 10 Relationship of dye class to fibre – reactive/direct/vat/sulphur for cellulose, disperse for polyester, acid for wool, silk and nylon, basic for acrylic
4. Basic chemicals	4.1 Caustic soda (sodium hydroxide) – scouring and mercerizing 4.2 Soda ash (sodium carbonate) – alkali for reactive fixation 4.3 Hydrogen peroxide – bleaching agent 4.4 Sodium hypochlorite and sodium chlorite – bleaching agents 4.5 Common salt (NaCl) and Glauber's salt (Na_2SO_4) – electrolyte 4.6 Acetic acid and formic acid – neutralization and pH control 4.7 Hydrochloric acid and sulphuric acid – souring and pH control 4.8 Sodium hydrosulphite (hydrosulphite) – reducing agent 4.9 Sodium sulphide – for sulphur dyes 4.10 Sodium bisulphite and metabisulphite – reducing/anti-chlor agents 4.11 Sodium bicarbonate and trisodium phosphate 4.12 Urea, sodium silicate and resist salt
5. Auxiliaries	5. 1 Wetting agent and penetrating agent 5. 2 Detergent and scouring agent 5. 3 Sequestering (chelating) agent 5. 4 Peroxide stabilizer 5. 5 Anti-creasing and lubricating agent 5. 6 Levelling, retarding and dispersing agent 5. 7 Enzymes – amylase (desizing), cellulase (bio-polishing), catalase (peroxide killer), pectinase (bio-scouring) 5. 8 Soaping and washing-off agent 5. 9 Anti-back-staining agent 5. 10 Fixing agent (cationic fixer)

	5. 11 Softener – cationic, non-ionic, anionic and silicone 5. 12 Anti-foaming agent 5. 13 Optical brightening agent (OBA) 5. 14 Resin, cross-linking agent and catalyst 5. 15 Buffer and pH-controlling agent
6. Pigments and printing chemicals	6.1 Pigment colours and binder 6.2 Thickeners – sodium alginate, guar gum, starch, CMC, synthetic and emulsion thickener 6.3 Urea, glycerine and humectants 6.4 Resist salt (sodium m-nitrobenzene sulphonate) 6.5 Fixer, catalyst and cross-linker 6.6 Discharge agent (decroline/rongalite) 6.7 Plastisol, puff, foil, glitter, high-density and rubber print chemicals 6.8 Anti-migrating agent and emulsifier 6.9 Softener and handle modifier for print
7. Machines used in wet processing	7. 1 Singeing machine (gas singeing) and desizing range 7. 2 J-box, kier and continuous bleaching range (CBR) 7. 3 Mercerizing machine (chain and chainless) 7. 4 Winch dyeing machine and jigger 7. 5 Soft-flow, overflow, jet and airflow (HTHP) dyeing machines 7. 6 Package (cone), hank and beam yarn dyeing machines 7. 7 Cold pad-batch (CPB) unit and padding mangle 7. 8 Continuous dyeing range (pad-dry-pad-steam, thermosol range) 7. 9 Sample/laboratory dyeing machine 7. 10 Rotary and flat-bed screen printing machines, hand printing table 7. 11 Digital (inkjet) textile printing machine and transfer printing press 7. 12 Loop steamer/ager and curing chamber 7. 13 Washing, soaping and open-width washing range 7. 14 Hydro-extractor, squeezer, slitting and de-watering machine 7. 15 Relax dryer, tumble dryer and stenter/heat setting machine 7. 16 Open-width and tubular compactor, sanforizing machine 7. 17 Calender, raising, sueding and shearing machine 7. 18 Fabric inspection, folding, rolling and packing machine 7. 19 Colour kitchen dispensing system and print paste kitchen 7. 20 Boiler, water treatment plant and effluent treatment plant (ETP)
8. Laboratory and measuring instruments	8.1 Analytical balance (0.001 g) and electronic balance (0.01 g) 8.2 pH meter, pH paper and buffer solutions

	8.3 Beaker, measuring cylinder, pipette, burette and volumetric flask 8.4 Thermometer and temperature probe 8.5 GSM cutter and GSM balance 8.6 Spectrophotometer with colour matching software 8.7 Light box/colour matching cabinet (D65, TL84, TL83, CWF, UV, fluorescent & incandescent A) 8.8 Crock meter (rubbing fastness tester) 8.9 Wash fastness tester (launder-o-meter/rota wash) and perspiration tester 8.10 Grey scale for colour change and staining 8.11 Shrinkage and spirality template with scale 8.12 Hydrometer/Baume meter for caustic concentration 8.13 Viscometer/flow cup for print paste 8.14 Titration set and laboratory glassware 8.15 Stopwatch, timer and measuring tape
9. Stages of the wet processing sequence	9.1 Grey fabric/yarn inspection, batching and stitching 9.2 Singeing (woven fabric) 9.3 Desizing (woven fabric) 9.4 Scouring 9.5 Bleaching 9.6 Mercerizing (where required) 9.7 Dyeing or printing 9.8 Washing-off, soaping and after-treatment 9.9 Neutralization and drying 9.10 Chemical finishing (padding and curing) 9.11 Mechanical finishing (stentering, compacting, calendering, sanforizing) 9.12 Final inspection, folding, packing and delivery
10. Process parameters	10.1 Material-to-liquor ratio (M:L) 10.2 Temperature and heating/cooling gradient (°C/min) 10.3 Process time and total cycle time 10.4 pH of the bath and of the fabric 10.5 Concentration of dyes and chemicals (g/l and % owf) 10.6 Shade percentage on weight of fabric (% owf) 10.7 Wet pick-up and expression percentage 10.8 Machine speed, dwell time and stenter residence time 10.9 Steam pressure, curing and heat setting temperature 10.10 Liquor circulation rate, nozzle pressure and rope speed 10.11 Water, steam and energy consumption per kg of fabric
11. Basic process calculations	11.1 Determination of fabric batch weight 11.2 Liquor quantity from the material-to-liquor ratio 11.3 Dye quantity from shade percentage (batch weight × shade% ÷ 100) 11.4 Chemical quantity from concentration in g/l (liquor volume × g/l) 11.5 Stock solution strength and dilution calculation

	11.6 Wet pick-up and add-on percentage 11.7 Percentage to gram and gram to percentage conversion 11.8 Production per shift and machine efficiency 11.9 Water and steam consumption per kg of fabric
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Curricular Content Guide:

1. Underpinning Knowledge	1. 1 Classification, sources, properties and identification of textile fibres. 1. 2 Yarn and fabric structures presented for wet processing and their processing implications. 1. 3 Classification of dyes, their properties, and their suitability for different fibres. 1. 4 Basic chemicals used in wet processing and the function of each. 1. 5 Auxiliaries and their functions in pre-treatment, dyeing, printing and finishing. 1. 6 Pigments, binders, thickeners and printing chemicals. 1. 7 Machines used at each stage of wet processing and their working principles. 1. 8 Laboratory and measuring instruments used in a dye house and their functions. 1. 9 Utilities – steam, water, gas, compressed air and electricity – and their role in wet processing. 1. 10 Process parameters and their influence on quality, cost and reproducibility. 1. 11 Water quality requirements and the effect of hard water on wet processing. 1. 12 Production documents, recipes, batch cards, SOPs and workplace terminology.
2. Underpinning Skills	2. 1 Identifying textile fibres by appearance, burning test and simple solubility test 2. 2 Identifying yarn and fabric structures and measuring GSM, width and construction 2. 3 Identifying dye classes and relating them to the fibre to be dyed 2. 4 Identifying chemicals and auxiliaries from labels and describing their functions 2. 5 Identifying machines and classifying them according to process stage 2. 6 Identifying and handling laboratory and measuring instruments 2. 7 Describing the wet processing sequence and the objective of each stage 2. 8 Interpreting recipes, process cards and machine programmes 2. 9 Testing and interpreting basic water quality parameters

	2. 10 Recording and reporting information using workplace formats
3. Underpinning Attitudes	3. 1 Commitment to occupational health and safety 3. 2 Promptness in carrying out activities 3. 3 Sincere and honest to duties 3. 4 Environmental concerns 3. 5 Eagerness to learn 3. 6 Tidiness and timeliness 3. 7 Respect for rights of peers and seniors in the workplace 3. 8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4. 1 Workplace (simulated or actual) 4. 2 Fibre sample board and fibre samples (natural, man-made, regenerated and blends) 4. 3 Samples of grey, pre-treated, dyed, printed and finished fabric 4. 4 Yarn samples in hank, cone and package form 4. 5 Sample sets of dyes, chemicals and auxiliaries with labels and safety data sheets 4. 6 Charts, models and flow diagrams of the wet processing sequence 4. 7 Laboratory glassware and measuring instruments 4. 8 Light box and shade cards 4. 9 Multimedia projector, laptop and audio-visual clips 4. 10 Learning manual, SOPs, sample batch cards, recipes and production formats

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1. 1 Interpreted textile wet processing and materials to be processed 1. 2 Identified dyes, chemicals and auxiliaries used in wet processing. 1. 3 Identified machines, tools and equipment used in wet processing. 1. 4 Described the wet processing sequence. 1. 5 Interpreted work instructions, recipes and production documents.
2. Methods of Assessment	Competency should be assessed by: 2. 1 Written test 2. 2 Practical demonstration 2. 3 Oral questioning 2. 4 Portfolio (Optional)

3. Context of Assessment	3. 1 Competency assessment must be done in an assessment/training centre or in an actual or simulated workplace after completion of the training. 3. 2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.
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Unit of Competency: APPLY DYES, CHEMICALS AND AUXILIARIES FOR WET PROCESSING	Nominal Duration: 20 Hrs.	Unit Code: SICIP-TEX-PPWP-02-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply dyes, chemicals and auxiliaries for wet processing. It specifically includes the tasks of preparing for chemical handling work, storing and handling dyes, chemicals and auxiliaries safely, weighing, dissolving and preparing dyes, chemicals and auxiliaries, and dispensing and dosing solutions to the machine.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare for chemical handling work	1. 1 <u>Personal protective equipment for chemical handling</u> is selected and used in accordance with safety guidelines. 1. 2 <u>Material Safety data sheet (MSDS) information</u> is obtained and interpreted before handling any chemical. 1. 3 <u>Colour lab equipment and instruments</u> are checked for cleanliness, calibration and readiness. 1. 4 Emergency facilities (eye wash station, safety shower, spill kit) are checked for availability and function. 1. 5 Ventilation and exhaust in the chemical handling area are confirmed as working.
2. Store and handle dyes, chemicals and auxiliaries safely	2. 1 Dyes, chemicals and auxiliaries are received, verified against the challan and identified from the label. 2. 2 <u>Storage requirements</u> for each chemical group are followed. 2. 3 Incompatible chemicals are segregated and stored separately. 2. 4 Containers are kept closed, labelled, undamaged and stored at the correct height. 2. 5 Chemicals are transported using appropriate material handling aids. 2. 6 First-in-first-out (FIFO) practice and shelf life are observed. 2. 7 Hazard symbols, safety signage and no-smoking rules are followed at all times.
3. Weigh, dissolve and prepare dyes, chemicals and auxiliaries	3. 1 Dyes and chemicals are weighed accurately using the appropriate weighing instrument. 3. 2 Dissolving and pasting methods appropriate to each dye class are applied. 3. 3 Correct water temperature and continuous stirring are maintained during dissolution.

	3. 4 Concentrated acids and alkalis are diluted following the correct sequence (acid added into water, never water into acid). 3. 5 <u>Stock solutions</u> are prepared, labelled and stored as per procedure.
4. Dispense and dose solutions to the machine	4. 1 Prepared solutions are dispensed to the correct machine and batch as per the process card. 4. 2 <u>Dosing methods</u> are applied according to the process requirement. 4. 3 Dosing sequence, rate and time are followed as specified in the process program. 4. 4 pH and temperature of the bath are checked before and after each addition. 4. 5 Operation of the automatic dispensing system is supported as per the level of authority. 4. 6 Deviations in dispensing, dosing or quantity are reported promptly to the authority.

Range of Variables:

Variable	Range (May include but not limited to)
1. Personal protective equipment for chemical handling	1. 1 Chemical splash goggles and face shield 1. 2 Chemical-resistant gloves (nitrile, rubber, PVC) 1. 3 Nose mask and chemical respirator/cartridge mask 1. 4 Rubber apron and chemical-resistant suit 1. 5 Rubber boots and anti-slip safety shoes 1. 6 Head cap and hair covering
2. Material Safety data sheet (MSDS) information	2. 1 Product and supplier identification 2. 2 Hazard identification, pictograms and signal words 2. 3 Composition and information on ingredients 2. 4 First aid measures 2. 5 Firefighting measures 2. 6 Accidental release and spill measures 2. 7 Handling and storage requirements 2. 8 Exposure controls and personal protection 2. 9 Physical and chemical properties 2. 10 Stability and reactivity/incompatibility 2. 11 Toxicological and ecological information 2. 12 Disposal considerations
3. Colour lab equipment and instruments	3. 1 Analytical balance and electronic balance 3. 2 Dissolving vessel with stirrer/agitator 3. 3 Measuring cylinder, beaker, pipette, funnel and jug 3. 4 Hot and cold water lines with mixing arrangement 3. 5 Scoop, spatula, stirring rod and sieve/strainer 3. 6 pH meter, pH paper and thermometer 3. 7 Dosing tank, dosing pump and dispensing line 3. 8 Automatic colour dispensing and dissolving system

	3. 9 Trolleys, containers, drums and storage racks 3. 10 Exhaust hood and ventilation arrangement
4. Storage requirements	4. 1 Dry, cool, well-ventilated and shaded store away from direct sunlight 4. 2 Separate designated areas for acids, alkalis and oxidizing agents 4. 3 Bunded/kerbed area with a spill containment sump for liquid chemicals 4. 4 Raised pallets and racks keeping containers off the wet floor 4. 5 Temperature-controlled storage for temperature-sensitive chemicals and enzymes 4. 6 Locked and secured storage for hazardous and restricted chemicals 4. 7 Clear labelling, hazard signage and updated inventory record 4. 8 Fire extinguishers, spill kit and eye wash within reach
5. Stock solutions	5.1 Concentration expressed in g/l and preparation date 5.2 Label showing name, strength, date of preparation and expiry 5.3 Shelf life and stability of the solution 5.4 Clean, covered and correctly sized storage container 5.5 Stirring/agitation before use to avoid settling 5.6 Separate stock and separate utensils for each chemical to avoid cross contamination
6. Dosing methods	6.1 Manual/bucket addition into the machine addition tank 6.2 Linear dosing at a constant rate 6.3 Progressive (exponential) dosing 6.4 Step-wise dosing in portions 6.5 Single-shot (all-in) addition/ 6.6 Automatic dosing pump and dosing controller 6.7 Dosing through the injection line with liquor circulation running

Curricular Content Guide:

1. Underpinning Knowledge	1. 1 Layout, function and discipline of the colour kitchen and chemical store. 1. 2 Classification of dyes and chemicals and their solubility behaviour. 1. 3 Chemical storage requirements, segregation of incompatible chemicals and shelf life. 1. 4 Correct dilution practice for concentrated acids and alkalis. 1. 5 Weighing instruments, their accuracy, calibration and correct use.
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	1. 6 Recipe calculations – liquor volume, dye quantity, chemical quantity and dilution. 1. 7 Dissolving and pasting methods for each dye class. 1. 8 Stock solution preparation, labelling, strength and shelf life. 1. 9 Dosing methods and the importance of the addition sequence on levelness. 1. 10 Effect of water quality and temperature on dissolution. 1. 11 Spill containment, neutralization and chemical waste disposal requirements. 1. 12 Chemical inventory management, traceability and consumption recording.
2. Underpinning Skills	2. 1 Interpreting recipes and process cards 2. 2 Checking and calibrating balances, pH meters and measuring instruments 2. 3 Receiving, verifying, labelling and storing chemicals correctly 2. 4 Performing recipe and dilution calculations correctly 2. 5 Weighing dyes and chemicals accurately with the correct instrument 2. 6 Pasting and dissolving dyes according to dye class 2. 7 Diluting concentrated acids and alkalis safely 2. 8 Preparing, labelling & storing stock solutions, and filtering & checking prepared solutions before dispensing 2. 9 Dispensing and dosing solutions to the correct machine and batch 2. 10 Containing and cleaning chemical spills using the spill kit 2. 11 Disposing of empty containers and residual chemicals correctly 2. 12 Completing chemical issue, consumption and inventory records
3. Underpinning Attitudes	3. 1 Commitment to occupational health and safety 3. 2 Promptness in carrying out activities 3. 3 Sincere and honest to duties 3. 4 Environmental concerns 3. 5 Eagerness to learn 3. 6 Tidiness and timeliness 3. 7 Respect for rights of peers and seniors in the workplace 3. 8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4. 1 Workplace (simulated or actual) 4. 2 Colour kitchen with dissolving vessels, stirrers and dosing arrangement 4. 3 Analytical balance (0.001 g) and electronic balance (0.01 g)

	4. 4 Laboratory glassware – beaker, measuring cylinder, pipette, funnel and flask, and pH meter, pH paper and thermometer 4. 5 Dyes of different classes, chemicals and auxiliaries in sample quantity 4. 6 Chemical storage rack and cabinet with segregation arrangement 4. 7 Eye wash station, emergency safety shower and chemical spill kit 4. 8 Drum trolley and material handling aids 4. 9 Hot and cold-water supply and exhaust/ventilation arrangement 4. 10 SOPs, sample recipes, process cards and recording formats 4. 11 Learning manual, calculator and audio-visual clips
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Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1. 1 Prepared for chemical handling work. 1. 2 Stored and handled dyes, chemicals and auxiliaries safely. 1. 3 Weighed, dissolved and prepared dyes, chemicals and auxiliaries. 1. 4 Dispensed and dosed solutions to the machine.
2. Methods of Assessment	Competency should be assessed by: 2. 1 Written test 2. 2 Practical demonstration 2. 3 Oral questioning 2. 4 Portfolio (Optional)
3. Context of Assessment	3. 1 Competency assessment must be done in an assessment/training centre or in an actual or simulated workplace after completion of the training. 3. 2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM YARN AND FABRIC PRE-TREATMENTS	Nominal Duration: 65 Hrs.	Unit Code: SICIP-TEX-PPWP-03-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform yarn and fabric pre-treatments. It specifically includes the tasks of preparing grey material and machines for pre-treatment, carrying out singeing and de-sizing operations, carrying out scouring, bleaching and bio-polishing operations, carrying out mercerizing operations, and checking pre-treatment quality and recording process data.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare grey material and machines for pre-treatment	1. 1 Work order, batch card and grey fabric inspection report are obtained and understood. 1. 2 Grey material is identified, weighed and batched according to lot, construction and shade group. 1. 3 Fabric is stitched/joined correctly and batching precautions are followed. 1. 4 <u>Pre-treatment machines</u> and accessories are checked for cleanliness and readiness. 1. 5 Steam, water, gas, air and electrical supply are confirmed as available and correct.
2. Carry out singeing and de-sizing operations	2. 1 <u>Singeing parameters</u> are set/verified according to fabric type and requirement. 2. 2 Fabric is passed through the singeing machine maintaining the correct flame position, intensity and speed. 2. 3 Singeing quality is checked and burning, scorching or damage to the fabric is prevented. 2. 4 Desizing method is selected and applied according to the size present in the fabric. 2. 5 Desizing liquor is prepared and applied at the specified concentration, temperature and pH. 2. 6 Adequate dwell/batching time is allowed for the desizing reaction to complete. 2. 7 Desized fabric is washed thoroughly and desizing efficiency is checked using the standard test.
3. Carry out scouring, bleaching and bio-polishing operations	3. 1 Scouring, bleaching and bio-polishing liquor is prepared as per the <u>recipe</u> and charged into the machine. 3. 2 <u>Chemicals and parameters</u> are applied as per the process requirement. 3. 3 The fabric is rinsed and neutralized to the required pH before the next process.

4. Carry out mercerizing operations	4.1 Caustic soda concentration is checked using a hydrometer/Baume meter and adjusted to the requirement. 4.2 Fabric is passed through the mercerizing machine at the specified tension, speed and temperature. 4.3 <u>Mercerizing parameters</u> are maintained throughout the operation. 4.4 Fabric is stabilized, washed and neutralized to the required pH. 4.5 Caustic recovery and reuse arrangements are followed where available. 4.6 Mercerizing quality is checked and reported. 4.7 Safety precautions for handling strong caustic solution are strictly followed at all times.
5. Check pre-treatment quality and record process data	5. 1 Samples are collected at the specified stage and <u>pre-treatment tests</u> are carried out. 5. 2 Results are compared against the standard and out-of-standard results are reported promptly. 5. 3 <u>Pre-treatment faults</u> are identified and their probable causes are reported. 5. 4 Corrective action is taken or reported as per the level of authority. 5. 5 Water, steam, chemical and energy consumption are recorded where required. 5. 6 Process and quality data are recorded in the prescribed format. 5. 7 Machine, filters and work area are cleaned after completion of the batch.

Range of Variables:

Variable	Range (May include but not limited to)
1. Pre-treatment machines	1. 1 Gas singeing machine with brushing and de-dusting unit 1. 2 Desizing pad–batch unit and J-box 1. 3 Kier, pressure vessel and continuous bleaching range (CBR) 1. 4 Winch, soft-flow, overflow and jet dyeing machines used for knit pre-treatment 1. 5 Jigger for woven fabric pre-treatment 1. 6 Chain and chainless mercerizing machine 1. 7 Washing range, squeezer and open-width washer 1. 8 Hydro-extractor and de-watering machine 1. 9 Enzyme treatment bath and dosing arrangement 1. 10 Yarn pre-treatment (package/hank) machine
2. Singeing parameters	2. 1 Burner type, number of burners and burner position

	2. 2 Flame intensity and gas pressure 2. 3 Flame position – tangential, normal (direct on fabric) or on water-cooled roller 2. 4 Fabric speed (generally 80–150 m/min depending on fabric) 2. 5 Distance between burner and fabric 2. 6 Cooling/quenching and dust extraction arrangement 2. 7 Singeing rating on the 1–5 scale (5 = best singed surface) 2. 8 Brushing and de-dusting before singeing
3. Recipe	3. 1 Names and concentrations of chemicals (caustic soda, soda ash, peroxide) 3. 2 Wetting, detergent, sequestering and stabilizing agent dosage 3. 3 Enzyme type, dosage and activity 3. 4 Optical brightening agent dosage for full white 3. 5 Material-to-liquor ratio and liquor volume 3. 6 Temperature, time and pH profile 3. 7 Number of rinse baths and rinse temperatures 3. 8 Neutralization and peroxide killer dosage
4. Chemicals and Parameters	4.1 Soda, acid, enzyme, detergent, hydrogen peroxide and auxiliaries 4.2 Temperature and run time 4.3 pH of the bath 4.4 Material-to-liquor ratio 4.5 Number and temperature of rinse baths
5. Mercerizing parameters	6.1 Caustic soda concentration and bath temperature (hot or cold mercerizing) 6.2 Impregnation/dwell time (generally 40–60 seconds) 6.3 Warp and weft tension and stretch applied 6.4 Machine speed and fabric width control 6.5 Stabilizing zone and hot washing zone conditions 6.6 Neutralization with acetic acid and final pH (generally 6.5–7.5) 6.7 Wetting agent addition for uniform caustic penetration 6.8 Residual caustic on the fabric after washing
6. Pre-treatment tests	7. 1 Absorbency/drop test (drop absorbed generally within 1–5 seconds) 7. 2 Whiteness index measurement using a spectrophotometer 7. 3 Residual size test using the TEGEWA violet scale 7. 4 pH measurement of the fabric extract and the bath 7. 5 Residual peroxide test using test strips 7. 6 Weight loss determination before and after processing 7. 7 Tensile, tearing and bursting strength test 7. 8 Barium activity number test for mercerized fabric

	7. 9 Cuprammonium fluidity test for chemical damage
7. Pre-treatment faults	8. 1 Uneven, patchy or poor absorbency 8. 2 Low, uneven or patchy whiteness 8. 3 Residual size, starch spots and stiff patches 8. 4 Oil, rust, grease and contamination stains remaining after scouring 8. 5 Pin holes and catalytic damage caused by iron particles 8. 6 Over-bleaching, fibre damage and loss of strength 8. 7 Crease marks, rope marks and abrasion marks 8. 8 Uneven singeing, singeing streaks and scorching 8. 9 Uneven mercerization and streaky lustre 8. 10 Residual alkali or peroxide left on the fabric 8. 11 Fabric distortion, width variation and edge curling

Curricular Content Guide:

1. Underpinning Knowledge	1. 1 Impurities present in cotton, synthetic and blended fabrics and the methods of removing them. 1. 2 Purpose, principle and machinery of singeing, and the singeing rating scale. 1. 3 Types of size used in weaving and the selection of the appropriate de-sizing method. 1. 4 Principle and objectives of scouring and the role of caustic soda, detergent and sequestering agents. 1. 5 Principle and objectives of bleaching and the chemistry of peroxide bleaching. 1. 6 Effect of pH, temperature, stabilizer and sequestering agent on peroxide bleaching. 1. 7 Combined pre-treatment processes and their advantages. 1. 8 Bio-scouring and bio-polishing using enzymes. 1. 9 Principle, objectives and machinery of mercerizing. 1. 10 Effect of tension, time and concentration on mercerizing quality. 1. 11 Common pre-treatment faults, their causes and remedies. 1. 12 Water, steam, chemical and energy consumption in pre-treatment and methods of reduction.
2. Underpinning Skills	2. 1 Interpreting batch cards, work orders and grey fabric inspection reports 2. 2 Weighing, batching and stitching grey fabric correctly 2. 3 Setting and verifying singeing parameters and assessing singeing quality 2. 4 Preparing and applying de-sizing liquor by the appropriate method 2. 5 Performing the TEGEWA/iodine test to check de-sizing efficiency

	2. 6 Preparing and charging scouring and bleaching liquor as per the recipe 2. 7 Monitoring temperature, time, pH, liquor level and fabric running condition 2. 8 Operating and monitoring the mercerizing machine under supervision 2. 9 Performing absorbency, whiteness, pH and strength tests 2. 10 Identifying pre-treatment faults and reporting their probable causes 2. 11 Recording process, quality and consumption data in prescribed formats
3. Underpinning Attitudes	3. 1 Commitment to occupational health and safety 3. 2 Promptness in carrying out activities 3. 3 Sincere and honest to duties 3. 4 Environmental concerns 3. 5 Eagerness to learn 3. 6 Tidiness and timeliness 3. 7 Respect for rights of peers and seniors in the workplace 3. 8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Singeing machine or demonstration/audio-visual facility 4.3 Winch, jigger, soft-flow or sample dyeing machine for pre-treatment practice 4.4 Mercerizing machine or laboratory mercerizing arrangement 4.5 Chemicals – caustic soda, soda ash, hydrogen peroxide, acids, enzymes and auxiliaries 4.6 pH meter, pH paper and peroxide test strips 4.7 TEGEWA violet scale and iodine solution 4.8 Electronic balance, glassware and measuring instruments 4.9 Spectrophotometer for whiteness measurement 4.10 Eye wash station, safety shower and spill kit 4.11 SOPs, batch cards, recipes and recording formats 4.12 Learning manual and audio-visual clips

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1. 1 Prepared grey material and machines for pre-treatment. 1. 2 Carried out singeing and de-sizing operations. 1. 3 Carried out scouring, bleaching and bio-polishing operations. 1. 4 Carried out mercerizing operations. 1. 5 Checked pre-treatment quality and recorded process data.
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2. Methods of Assessment	Competency should be assessed by: 2. 1 Written test 2. 2 Practical demonstration 2. 3 Oral questioning 2. 4 Portfolio (Optional)
3. Context of Assessment	3. 1 Competency assessment must be done in an assessment/training centre or in an actual or simulated workplace after completion of the training. 3. 2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM DYEING OPERATIONS	Nominal Duration: 90 Hrs.	Unit Code: SICIP-TEX-PPWP-04-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform dyeing operations. It specifically includes the tasks of preparing the batch and dyeing machine, preparing and charging the dye bath, carrying out the dyeing operation, performing after-treatment, washing-off and soaping, and checking shade and dyeing quality and recording data.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare the batch and dyeing machine	1. 1 Batch card, <u>recipe</u> and <u>lab dip/shade approval</u> are obtained and understood. 1. 2 <u>Dyeing machines</u> are selected/confirmed according to material form, batch weight and shade. 1. 3 Machine, <u>machine parts</u> , nozzles and filters are checked for cleanliness and residual colour. 1. 4 Pre-treated material is checked for absorbency, pH and residual peroxide before loading. 1. 5 Material is loaded correctly maintaining rope length and avoiding entanglement. 1. 6 Water quality, quantity and the material-to-liquor ratio are confirmed as per the requirement.
2. Prepare and charge the dye bath	2. 1 Dyes, chemicals and auxiliaries are received from the colour kitchen and verified against the recipe. 2. 2 Levelling, sequestering and auxiliary agents are added following the specified addition sequence. 2. 3 <u>pH of the dye bath</u> is set and verified using a pH meter or pH paper. 2. 4 Dye solution is charged into the machine following the specified dosing profile. 2. 5 Salt/electrolyte and alkali are added at the specified time, rate and temperature. 2. 6 Bath temperature is raised or lowered following the specified gradient. 2. 7 Uniform circulation of liquor and material is confirmed before and during dyeing.
3. Carry out the dyeing operation	3. 1 The machine programme is selected and started following standard operating procedures. 3. 2 <u>Dyeing parameters</u> are monitored and maintained throughout the cycle. 3. 3 <u>Dyeing methods</u> are applied according to the dye class and material form. 3. 4 Fabric running condition, rope speed, liquor level and foam are observed continuously.

	3. 5 Samples are drawn at the specified stage for shade checking. 3. 6 Machine stoppages and process abnormalities are attended or reported according to the level of authority. 3. 7 Shade correction or dye addition is carried out under supervision when instructed.
4. Perform after-treatment, washing-off and soaping	4.1 The bath is drained and the material is rinsed following the specified sequence. 4.2 Neutralization is carried out to bring the material to the required pH. 4.3 Soaping and washing-off is performed at the specified temperature and time to remove unfixed dye. 4.4 After-treatments are applied according to the dye class and fastness requirement. 4.5 Softener or finishing chemical is applied in the machine where specified. 4.6 Material is unloaded, de-watered/hydro-extracted and delivered to the next process without damage. 4.7 The machine, filter and bath are cleaned to prevent shade contamination of the next batch.
5. Check shade and dyeing quality and record data	5. 1 Dyed samples are dried, conditioned and compared with the approved standard under the specified light sources. 5. 2 Shade assessment is carried out and the result is reported to the authority. 5. 3 <u>Fastness properties</u> are tested or samples are submitted for testing as per procedure. 5. 4 <u>Dyeing faults</u> are identified and their probable causes are reported. 5. 5 Re-processing and shade correction requirements are reported and recorded. 5. 6 Right-first-time (RFT) percentage, water, steam and energy consumption are recorded where required. 5. 7 Production and quality data are recorded in the prescribed format.

Range of Variables:

Variable	Range (May include but not limited to)
1. Recipe	1. 1 Shade name, reference and depth (% owf) 1. 2 Dye class and individual dye components with percentages 1. 3 Salt/electrolyte concentration (g/l) 1. 4 Auxiliary names and dosage (g/l or % owf) 1. 5 Material-to-liquor ratio and liquor volume 1. 6 Temperature, time and pH 1. 7 Dosing profile, addition sequence and dosing time
2. Lab dip/shade approval	2. 1 Lab dip sample and swatch with recipe

	2. 2 Buyer-approved standard, pantone or colour reference 2. 3 Light sources for approval – D65 (daylight), TL84, CWF, incandescent A and UV 2. 4 Colour difference (DE) tolerance allowed by the buyer 2. 5 Continuity card and lab-to-bulk correlation 2. 6 Approval signature, date and validity
3. Dyeing machines	3. 1 Winch dyeing machine 3. 2 Jigger dyeing machine 3. 3 Soft-flow, overflow and jet (HTHP) dyeing machines 3. 4 Airflow dyeing machine 3. 5 Package (cone) yarn dyeing machine 3. 6 Hank yarn dyeing machine 3. 7 Beam dyeing machine 3. 8 Cold pad-batch (CPB) unit 3. 9 Continuous dyeing range (pad-dry-pad-steam, thermosol range) 3. 10 Sample/laboratory dyeing machine (infrared, glycerine bath) 3. 11 Garment dyeing machine
4. Machine parts	4.1 Main vessel, kier and nozzle 4.2 Winch reel, lifter roller and plaiter 4.3 Circulation pump and heat exchanger 4.4 Addition/dosing tank and injection line 4.5 Filter, lint trap and strainer 4.6 Temperature, pressure and level sensors 4.7 Sampling device and sampling cock 4.8 Fill, drain and overflow valves 4.9 Control panel/PLC with programme memory 4.10 Loading and unloading arrangement 4.11 Steam, cooling water and air connections
5. pH of the dye bath	5. 1 Reactive dyeing – alkaline, generally pH 10.5–11.5 at fixation 5. 2 Disperse dyeing – acidic, generally pH 4.5–5.5 5. 3 Acid dyeing on nylon, wool and silk – generally pH 4–6 5. 4 Direct dyeing – neutral to slightly alkaline 5. 5 Vat and sulphur dyeing – strongly alkaline 5. 6 Basic dyeing on acrylic – generally pH 4–4.5 5. 7 Adjustment using acetic acid, soda ash or buffer 5. 8 Measurement using a pH meter and pH paper
6. Dyeing parameters	6.1 Temperature and heating/cooling gradient (°C/min) 6.2 Dyeing time at the peak temperature 6.3 pH of the bath 6.4 Liquor circulation rate and nozzle pressure 6.5 Rope speed and cycle time 6.6 Material-to-liquor ratio 6.7 Dye and chemical concentration

	6.8 Machine pressure for high-temperature dyeing 6.9 Exhaustion percentage and fixation percentage
7. Dyeing methods	7.1 Exhaust (batch) dyeing 7.2 Continuous and semi-continuous dyeing 7.3 Cold pad-batch (CPB) dyeing 7.4 Pad-dry-pad-steam method 7.5 Thermosol dyeing of polyester 7.6 All-in, migration and isothermal methods for reactive dyes 7.7 High-temperature (generally 130°C) dyeing of polyester 7.8 Carrier dyeing of polyester at boiling temperature 7.9 One-bath and two-bath dyeing of polyester-cotton blends 7.10 Vat and sulphur dyeing with reduction, exhaustion and oxidation 7.11 Rope-form and open-width dyeing 7.12 Package, hank and beam dyeing of yarn
8. Fastness properties	8. 1 Colour fastness to washing 8. 2 Colour fastness to rubbing (dry and wet crocking) 8. 3 Colour fastness to perspiration (acid and alkaline) 8. 4 Colour fastness to light 8. 5 Colour fastness to water and sea water 8. 6 Colour fastness to hot pressing and sublimation 8. 7 Colour fastness to chlorinated water 8. 8 Grey scale rating (1–5) for colour change and staining 8. 9 Phenolic yellowing and colour fastness to saliva for children's wear
9. Dyeing faults	9.1 Uneven, patchy and unlevel dyeing 9.2 Shade variation side-centre-side and head-to-tail 9.3 Batch-to-batch and lot-to-lot shade variation 9.4 Crease marks, rope marks and abrasion marks 9.5 Dye spots, stains, specks and undissolved dye particles 9.6 Poor washing fastness, bleeding and colour bleeding on white 9.7 Poor rubbing fastness 9.8 Barre, streaks and stripes appearing after dyeing 9.9 Tailing in continuous dyeing 9.10 Back staining on blends and on white ground 9.11 Metamerism between the sample and the standard 9.12 Pin holes, fabric damage and strength loss 9.13 Uneven pick-up in pad dyeing and listing across the width

Curricular Content Guide:

1. Underpinning Knowledge	1. 1 Classification of dyes and the selection of a dye class for a given fibre and end use. 1. 2 Construction, working principle and application area of each type of dyeing machine. 1. 3 Material-to-liquor ratio and its effect on cost, levelness and reproducibility. 1. 4 Importance of pH control at each stage of dyeing for each dye class. 1. 5 Dyeing methods – exhaust, continuous, semi-continuous, CPB and thermosol. 1. 6 Vat, sulphur and indigo dyeing – reduction, exhaustion and oxidation. 1. 7 Washing-off, soaping and the removal of hydrolysed dye. 1. 8 Lab dip development, shade matching and the lab-to-bulk correlation. 1. 9 Light sources, metamerism and the principle of visual and instrumental shade assessment. 1. 10 Common dyeing faults, their causes and remedies. 1. 11 Water, salt, steam and energy consumption in dyeing and methods of reduction. 1. 12 Documentation, traceability and record-keeping requirements of the dye house.
2. Underpinning Skills	2. 1 Interpreting batch cards, recipes and lab dip approval documents 2. 2 Checking pre-treated material for absorbency, pH and residual peroxide 2. 3 Adding auxiliaries in the correct sequence 2. 4 Setting and verifying the pH of the dye bath 2. 5 Charging dye solution following the specified dosing profile 2. 6 Adding salt and alkali at the correct time, rate and temperature 2. 7 Monitoring temperature, pressure, pH, liquor level and circulation during dyeing 2. 8 Drawing samples safely during the dyeing cycle 2. 9 Applying after-treatments appropriate to the dye class 2. 10 Performing rubbing and wash fastness tests 2. 11 Identifying dyeing faults and reporting their probable causes 2. 12 Recording production, quality and consumption data in prescribed formats
3. Underpinning Attitudes	3. 1 Commitment to occupational health and safety 3. 2 Promptness in carrying out activities 3. 3 Sincere and honest to duties 3. 4 Environmental concerns 3. 5 Eagerness to learn

	3. 6 Tidiness and timeliness 3. 7 Respect for rights of peers and seniors in the workplace 3. 8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Winch, jigger, soft-flow or sample dyeing machine with accessories 4.3 Yarn (package/hank) dyeing sample machine 4.4 Laboratory padding mangle and sample dryer 4.5 Sample/laboratory dyeing machine for lab dips 4.6 Dyes of different classes (reactive, disperse, direct, vat, sulphur, acid) 4.7 pH meter, pH paper and thermometer 4.8 Spectrophotometer with colour matching software 4.9 Light box/colour matching cabinet (D65, TL84, CWF, UV, A) 4.10 Crock meter, wash fastness tester and grey scale 4.11 Eye wash station, safety shower and spill kit 4.12 SOPs, batch cards, recipes, shade cards and recording formats 4.13 Learning manual, calculator and audio-visual clips

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1. 1 Prepared the batch and dyeing machine. 1. 2 Prepared and charged the dye bath. 1. 3 Carried out the dyeing operation. 1. 4 Performed after-treatment, washing-off and soaping. 1. 5 Checked shade and dyeing quality and recorded data.
2. Methods of Assessment	Competency should be assessed by: 2. 1 Written test 2. 2 Practical demonstration 2. 3 Oral questioning 2. 4 Portfolio (Optional)
3. Context of Assessment	3. 1 Competency assessment must be done in an assessment/training centre or in an actual or simulated workplace after completion of the training. 3. 2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM PRINTING OPERATIONS	Nominal Duration: 50 Hrs.	Unit Code: SICIP-TEX-PPWP-05-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform printing operations. It specifically includes the tasks of preparing for printing operations, preparing print paste, carrying out printing operations, carrying out drying, fixation and washing-off after printing, and checking print quality and recording data.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare for printing operations	1. 1 Work order, design and strike-off approval and print specification are obtained and understood. 1. 2 Printing machines and accessories are checked for cleanliness and readiness. 1. 3 Screens are checked for design, mesh, repeat, registration and damage. 1. 4 Prepared fabric is checked for absorbency, pH, shade and freedom from defects. 1. 5 Printing table, blanket and gumming arrangement are checked and prepared as required.
2. Prepare print paste	2. 1 <u>Thickener</u> is prepared or selected according to the printing style and fabric. 2. 2 Print paste ingredients are weighed and mixed as per the print paste recipe. 2. 3 Paste viscosity is checked and adjusted as per the requirement. 2. 4 Paste is stirred, filtered and kept free from lumps and air bubbles. 2. 5 <u>Printing styles</u> are prepared according to the design requirement. 2. 6 Paste is labelled with design, colour and batch details and stored appropriately. 2. 7 Residual and returned paste is handled, reused or disposed of following workplace procedure.
3. Carry out printing operations	3. 1 Screens are mounted and registration and repeat are set correctly. 3. 2 Printing parameters are set and verified as per the requirement. 3. 3 Fabric is fed and gummed to the blanket without crease, distortion or slippage. 3. 4 The machine is started and stopped following standard operating procedures. 3. 5 The first print/strike-off is checked and approved before running the full lot.

	3. 6 Paste level in the screen is maintained and squeegee movement is monitored continuously. 3. 7 Printing faults occurring during running are identified and attended or reported promptly. 3. 8 Blanket washing and screen cleaning are carried out as per the schedule.
4. Carry out drying, fixation and washing-off after printing	4.1 Printed fabric is dried at the specified temperature without disturbing or smudging the print. 4.2 Fixation methods appropriate to the dye or pigment class are applied. 4.3 Steaming or curing time and temperature are maintained as specified. 4.4 Washing-off after printing is carried out to remove thickener and unfixed dye. 4.5 Anti-staining and soaping agents are used as specified to prevent staining of the ground. 4.6 Washed fabric is dried and delivered to finishing without damage. 4.7 Effluent from printing and washing is directed to the collection system as per procedure.
5. Check print quality and record data	5.1 Printed fabric is inspected against the approved strike-off and the specification. 5.2 Print quality parameters are checked and recorded. 5.3 Fastness of the print is tested or samples are submitted for testing as per procedure. 5.4 Corrective action is taken or reported as per the level of authority. 5.5 Paste consumption, wastage and production data are recorded in the prescribed format. 5.6 Screens, squeegees, vessels and the work area are cleaned after completion of work.

Range of Variables:

Variable	Range (May include but not limited to)
1. Thickener	1. 1 Sodium alginate – for reactive printing 1. 2 Guar gum and its derivatives 1. 3 Starch and British gum – for selected applications 1. 4 Carboxymethyl cellulose (CMC) 1. 5 Synthetic/acrylic thickener – for pigment printing 1. 6 Emulsion thickener and half-emulsion thickener 1. 7 Preparation of stock thickening, soaking time and homogenization 1. 8 Selection based on dye class, fabric and printing style
2. Printing styles	2. 1 Direct printing 2. 2 Discharge printing – white discharge and coloured discharge

	2. 3 Resist printing 2. 4 Pigment printing 2. 5 Reactive printing on cellulose 2. 6 Disperse printing on polyester 2. 7 Burn-out (devöre) printing 2. 8 Foil, glitter, puff and high-density printing 2. 9 Rubber and plastisol printing 2. 10 Transfer printing 2. 11 Digital printing 2. 12 Flock printing
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Curricular Content Guide:

1. Underpinning Knowledge	1. 1 Printing styles – direct, discharge, resist, pigment and special effect printing. 1. 2 Construction and working principle of rotary, flat-bed, hand table and digital printing machines. 1. 3 Thickeners – types, properties, preparation and selection. 1. 4 Print paste recipe formulation for each dye class and printing style. 1. 5 Print paste viscosity, its measurement and its effect on print quality. 1. 6 Squeegee types, pressure, angle and their effect on the printed result. 1. 7 Drying and fixation methods for reactive, disperse and pigment prints. 1. 8 Washing-off of printed fabric and prevention of ground staining. 1. 9 Print quality parameters, fastness requirements and buyer specifications. 1. 10 Common printing faults, their causes and remedies. 1. 11 Paste wastage control, paste reuse and effluent load from printing. 1. 12 Documentation and record-keeping requirements of the printing department.
2. Underpinning Skills	2. 1 Checking screens for design, mesh, repeat and damage 2. 2 Checking printing machines, blanket and gumming arrangement before start-up 2. 3 Preparing stock thickening and print paste as per the recipe 2. 4 Measuring and adjusting paste viscosity 2. 5 Setting squeegee pressure, position and machine speed 2. 6 Feeding and gumming fabric to the blanket without distortion 2. 7 Checking and approving the first print before bulk running 2. 8 Maintaining paste level and monitoring the print during running

	2. 9 Identifying printing faults during running and taking corrective action 2. 10 Carrying out drying, steaming and curing at the specified conditions 2. 11 Performing rubbing and wash fastness tests on printed samples 2. 12 Recording paste consumption, production and quality data
3. Underpinning Attitudes	3. 1 Commitment to occupational health and safety 3. 2 Promptness in carrying out activities 3. 3 Sincere and honest to duties 3. 4 Environmental concerns 3. 5 Eagerness to learn 3. 6 Tidiness and timeliness 3. 7 Respect for rights of peers and seniors in the workplace 3. 8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Flat-bed screen printing table with squeegee set 4.3 Screen frames, screens and squeegees of assorted mesh 4.4 Print paste mixer, stirrer and homogenizer 4.5 Laboratory/mini steamer or ager and curing chamber 4.6 Pigments, reactive and disperse dyes for printing 4.7 Thickeners, binders, fixers, urea, resist salt and auxiliaries 4.8 Viscometer or flow cup 4.9 Light box, clock meter and wash fastness tester 4.10 SOPs, design sheets, strike-offs, paste recipes and recording formats 4.11 Learning manual and audio-visual clips

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1. 1 Prepared for printing operations. 1. 2 Prepared print paste. 1. 3 Carried out printing operations. 1. 4 Carried out drying, fixation and washing-off after printing. 1. 5 Checked print quality and recorded data.
2. Methods of Assessment	Competency should be assessed by: 2. 1 Written test 2. 2 Practical demonstration 2. 3 Oral questioning 2. 4 Portfolio (Optional)

3. Context of Assessment	3. 1 Competency assessment must be done in an assessment/training centre or in an actual or simulated workplace after completion of the training. 3. 2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.
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Unit of Competency: PERFORM FINISHING OPERATIONS	Nominal Duration: 50 Hrs.	Unit Code: SICIP-TEX-PPWP-06-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform finishing operations. It specifically includes the tasks of preparing the fabric and finishing machines for finishing, carrying out mechanical finishing operations, carrying out chemical finishing operations, performing stentering, heat setting and dimensional stabilization, and checking finished fabric quality and preparing for delivery.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare the fabric and finishing machines for finishing	1. 1 Work order, finishing specification and buyer requirement are obtained and understood. 1. 2 Dyed or printed fabric is checked for shade, pH, moisture content and defects before finishing. 1. 3 <u>Finishing machines</u> and accessories are checked for cleanliness and readiness. 1. 4 Fabric is slit, de-watered, opened and centred as required for open-width processing.
2. Carry out mechanical finishing operations	2. 1 <u>Mechanical finishing processes</u> are carried out as specified in the finishing specification. 2. 2 Machine settings for mechanical finishing are set and verified as per the requirement. 2. 3 Fabric tension, overfeed and width are controlled during the operation. 2. 4 Fabric surface, handle and appearance are observed continuously during running. 2. 5 Compaction and shrinkage control requirements are achieved as specified. 2. 6 Faults arising during mechanical finishing are attended or reported promptly. 2. 7 Waste, fly and lint are collected and disposed of as per workplace procedure.
3. Carry out chemical finishing operations	3. 1 <u>Finishing chemicals</u> are selected and the finishing liquor is prepared as per the recipe. 3. 2 Liquor concentration, pH and temperature are checked before application. 3. 3 Application methods are used according to the finish and the fabric requirement. 3. 4 Pick-up/expression percentage is checked and controlled by adjusting padder pressure. 3. 5 Uniform application across the width and along the length of the fabric is ensured.

	3. 6 <u>Types of chemical finish</u> are applied according to the buyer requirement. 3. 7 Drying and curing conditions appropriate to the finish are maintained.
4. Perform stentering, heat setting and dimensional stabilization	4.1 Stenter parameters are set and verified as per fabric type and finishing requirement. 4.2 Fabric is pinned or clipped correctly on the chain without selvedge damage. 4.3 Heat setting conditions are maintained for synthetic and elastane-containing fabrics. 4.4 Width, overfeed and residual moisture are controlled to achieve the specified dimensional requirements. 4.5 Weft straightening and bow/skew correction devices are used as required. 4.6 Exhaust, air circulation and chamber temperature are monitored during operation. 4.7 Stenter faults are identified and attended or reported promptly.
5. Check finished fabric quality and prepare for delivery	5.1 Finished fabric is inspected on the inspection machine and defects are marked. 5.2 Finished fabric quality parameters are checked against the specification. 5.3 Finishing faults are identified and their probable causes are reported. 5.4 Fabric is folded or rolled, measured, weighed and identified with lot and roll details. 5.5 Packing and labelling are completed as per the specification. 5.6 Fabric is stored and dispatched from the designated area following workplace procedure. 5.7 Production, quality and delivery data are recorded in the prescribed format.

Range of Variables:

Variable	Range (May include but not limited to)
1. Finishing machines	1. 1 Stenter (pin or clip type, gas or thermal oil heated) 1. 2 Relax dryer and tumble dryer 1. 3 Open-width and tubular compactor 1. 4 Sanforizing (rubber belt shrinking) machine 1. 5 Calender – friction, chasing and embossing 1. 6 Raising/napping machine 1. 7 Sueding/peaching (emery) machine 1. 8 Shearing/cropping machine 1. 9 Hydro-extractor, squeezer and balloon padder 1. 10 Padding mangle (two-bowl and three-bowl) 1. 11 Slitting, de-watering and centring machine

	1. 12 Fabric inspection machine 1. 13 Folding, rolling, weighing and packing machine 1. 14 Coating and lamination machine where available
2. Mechanical finishing processes	2. 1 Drying – relax drying and tumble drying 2. 2 Compacting – open-width and tubular 2. 3 Sanforizing (compressive shrinkage) 2. 4 Calendering – friction, chasing, embossing and schreiner 2. 5 Raising and napping 2. 6 Sueding and peaching 2. 7 Shearing and cropping 2. 8 Brushing and polishing 2. 9 Decatizing 2. 10 Tumbling and softening for hand feel 2. 11 Slitting, opening and centring of tubular fabric
3. Finishing chemicals	3. 1 Softener – cationic, non-ionic, anionic and silicone (micro and macro emulsion) 3. 2 Resin and cross-linking agent for wrinkle-free and easy-care finish 3. 3 Optical brightening agent for white fabric 3. 4 Anti-pilling agent and enzyme (bio-polishing) 3. 5 Hydrophilic and moisture-management/wicking agent 3. 6 Water and oil repellent (fluorocarbon and non-fluorocarbon) 3. 7 Flame retardant 3. 8 Anti-microbial and anti-odour agent 3. 9 UV protection agent 3. 10 Stiffening and filling agent 3. 11 Anti-static agent 3. 12 Catalyst, pH buffer and wetting agent
4. Types of chemical finish	4.1 Softening finish 4.2 Wrinkle-free, easy-care and durable-press finish 4.3 Water repellent and water proof finish 4.4 Oil and stain repellent finish 4.5 Flame retardant finish 4.6 Anti-microbial and anti-odour finish 4.7 Anti-pilling finish 4.8 Hydrophilic and moisture-management finish 4.9 UV protective finish 4.10 Anti-static finish 4.11 Enzyme (bio-polishing) finish 4.12 Stiff and crisp finish 4.13 Peach and suede handle finish

Curricular Content Guide:

1. Underpinning Knowledge	1. 1 Objectives and classification of finishing – mechanical, chemical and combined. 1. 2 Sequence of finishing operations for knit and woven fabrics. 1. 3 Construction and working principle of the stenter, compactor, sanforizer, calender, raising and sueding machines. 1. 4 Function of overfeed, tension and width control in finishing. 1. 5 Spirality and torque in knit fabric, their causes and control. 1. 6 Finishing chemicals, their functions, application methods and precautions. 1. 7 Pad–dry–cure process and the role of catalyst and curing conditions. 1. 8 Fabric inspection systems, particularly the four-point system, and defect scoring. 1. 9 Finished fabric quality parameters, test methods and tolerances. 1. 10 Common finishing faults, their causes and remedies. 1. 11 Packing, labelling, storage and dispatch requirements.
2. Underpinning Skills	2. 1 Checking dyed and printed fabric for shade, pH, moisture and defects 2. 2 Slitting, de-watering, opening and centring tubular fabric 2. 3 Setting machine speed, overfeed, tension and width 2. 4 Achieving specified shrinkage and spirality through overfeed and compaction control 2. 5 Preparing finishing liquor and checking concentration, pH and temperature 2. 6 Setting padder pressure and controlling wet pick-up 2. 7 Setting stenter zone temperatures, speed, overfeed and width 2. 8 Pinning and clipping fabric correctly on the stenter chain 2. 9 Using weft straightening and bow/skew correction devices 2. 10 Inspecting fabric on the inspection machine and applying the four-point system 2. 11 Identifying finishing faults and reporting their probable causes 2. 12 Folding, rolling, weighing, labelling and packing finished fabric
3. Underpinning Attitudes	3. 1 Commitment to occupational health and safety 3. 2 Promptness in carrying out activities 3. 3 Sincere and honest to duties 3. 4 Environmental concerns 3. 5 Eagerness to learn

	3. 6 Tidiness and timeliness 3. 7 Respect for rights of peers and seniors in the workplace 3. 8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Laboratory or mini stenter cum curing machine 4.3 Calender, raising or sueding machine 4.4 Hydro-extractor and de-watering arrangement 4.5 Slitting and centring arrangement for tubular fabric 4.6 Fabric inspection machine with lighting arrangement 4.7 Finishing chemicals – softeners, resins, repellents, anti-pilling and OBA 4.8 Shrinkage and spirality templates 4.9 pH meter, thermometer and moisture meter 4.10 Packing materials – poly bag, carton, sticker and strapping 4.11 SOPs, finishing specifications, recipes and recording formats 4.12 Learning manual and audio-visual clips

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1. 1 Prepared the fabric and finishing machines for finishing. 1. 2 Carried out mechanical finishing operations. 1. 3 Carried out chemical finishing operations. 1. 4 Performed stentering, heat setting and dimensional stabilization. 1. 5 Checked finished fabric quality and prepared for delivery.
2. Methods of Assessment	Competency should be assessed by: 2. 1 Written test 2. 2 Practical demonstration 2. 3 Oral questioning 2. 4 Portfolio (Optional)
3. Context of Assessment	3. 1 Competency assessment must be done in an assessment/training centre or in an actual or simulated workplace after completion of the training. 3. 2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM IN-PROCESS QUALITY CHECKS AND ROUTINE MACHINE CARE	Nominal Duration: 15 Hrs.	Unit Code: SICIP-TEX-PPWP-07-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform in-process quality checks and routine machine care. It specifically includes the tasks of performing in-process quality checks and basic tests, recording and reporting production and quality data, and maintaining housekeeping and workplace conditions.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Perform in-process quality checks and basic tests	1. The sampling procedure is followed at the specified stages and intervals. 2. Basic quality tests are performed using the appropriate instruments. 3. Quality parameters are checked against the standard and the buyer specification. 4. Out-of-standard results are identified and reported promptly. 5. Test samples are labelled, conditioned, handled and preserved as per procedure. 6. Testing instruments are handled carefully, kept clean and checked for calibration.
2. Record and report production and quality data	1. Production, consumption and quality data are recorded in the prescribed formats. 2. Basic calculations are performed correctly where required. 3. Reports are prepared and submitted to the authority within the required time. 4. Abnormalities, deviations and breakdowns are reported through the correct reporting channel. 5. Batch traceability from grey material to delivery is maintained through correct records. 6. Records are kept clean, legible, complete and traceable.
3. Maintain housekeeping and workplace conditions	2. <u>Workplace conditions</u> are checked and abnormalities are reported. 3. Wet floors, spillage and drainage are attended promptly to prevent slip hazards. 4. Materials, chemicals, tools and waste are stored in designated places. 5. Walkways, machine aisles and emergency exits are kept clear at all times. 6. Waste is segregated and disposed of as per the environmental procedure.

	7. The work area is handed over to the next shift in a clean and safe condition.
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Range of Variables:

Variable	Range (May include but not limited to)
1. Workplace conditions	1. 1 Floor wetness, drainage condition and slip risk 1. 2 Ventilation, exhaust and air circulation 1. 3 Chemical odour and fume level 1. 4 Cleanliness of the floor, machine surroundings and emergency exit

Curricular Content Guide:

1. Underpinning Knowledge	1. 1 Importance of in-process quality control in wet processing and the cost of re-processing. 1. 2 Sampling principles, frequency, conditioning and standard testing atmosphere. 1. 3 Basic testing instruments, their use, care and calibration requirements. 1. 4 Quality parameters, standards and permissible tolerances at each process stage. 1. 5 Interpretation of test results and identification of out-of-standard conditions. 1. 6 Four-point inspection system and scoring fabric 1. 7 Documentation, batch traceability, record keeping and reporting channels of the mill. 1. 8 Importance of machine cleaning in preventing shade contamination. 1. 9 Concept and application of 5S and housekeeping in a wet processing unit. 1. 10 Effect of wet floors, humidity, heat, chemical fumes and lighting on safety and health. 1. 11 Waste segregation and environmentally sound disposal practices.
2. Underpinning Skills	2. 1 Performing basic quality tests using appropriate instruments 2. 2 Applying the four-point inspection system and scoring fabric 2. 3 Recording production, consumption and quality data in prescribed formats 2. 4 Carrying out routine cleaning, descaling and lubrication as per schedule 2. 5 Cleaning machine vessels, nozzles and filters to prevent shade contamination

	2. 6 Identifying worn parts, leakages and abnormal machine conditions and reporting them 2. 7 Applying lockout–tagout and safety procedures during maintenance 2. 8 Selecting, using and storing hand tools and instruments correctly 2. 9 Applying 5S and housekeeping practices in the work area 2. 10 Checking and reporting workplace conditions and slip hazards 2. 11 Segregating and disposing of waste as per environmental procedure
3. Underpinning Attitudes	3. 1 Commitment to occupational health and safety 3. 2 Promptness in carrying out activities 3. 3 Sincere and honest to duties 3. 4 Environmental concerns 3. 5 Eagerness to learn 3. 6 Tidiness and timeliness 3. 7 Respect for rights of peers and seniors in the workplace 3. 8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4. 1 Workplace (simulated or actual) 4. 2 Spectrophotometer and light box/colour matching cabinet 4. 3 Crock meter, wash fastness tester, perspiration tester and grey scale 4. 4 Shrinkage and spirality templates 4. 5 Peroxide test strips, TEGEWA violet scale and iodine solution 4. 6 Fabric inspection machine with four-point system charts 4. 7 Lubrication chart, maintenance checklist and cleaning schedule 4. 8 Lockout–tagout kit, permit-to-work forms and safety signage 4. 9 Colour-coded waste bins and designated storage arrangement 4. 10 Learning manual, calculator and audio-visual clips

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1. 1 Performed in-process quality checks and basic tests. 1. 2 Recorded and reported production and quality data. 1. 3 Maintained housekeeping and workplace conditions. 1. 4 Carried out required maintenance and routine machine care
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2. Methods of Assessment	Competency should be assessed by: 2. 1 Written test 2. 2 Practical demonstration 2. 3 Oral questioning 2. 4 Portfolio (Optional)
3. Context of Assessment	3. 1 Competency assessment must be done in an assessment/training centre or in an actual or simulated workplace after completion of the training. 3. 2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

End of the Competency Standard

WORKSHOP / LAB FACILITY STANDARD

Course Name:	Production Process in Wet Processing
Number of Trainees:	25

Course-wise Training Space (Theoretical Classroom, Workshop/Lab/Classroom cum Workshop):

1. Classroom – 500 sft (46 sqm)
2. Workshop/Lab – 6,000 sft (557 sqm), OR
3. Classroom cum workshop – 6,500 sft (603 sqm)

The workshop/lab must have an impervious, anti-slip and properly drained floor, adequate ventilation and exhaust, a separate and ventilated chemical store, a separate colour kitchen area, and a safe steam, water and effluent connection.

Major Training Equipment and Training Facilities:

S.N.	Major Equipment and Training Facilities	Required Quantity
1.	Computer/Laptop	2
2.	Multimedia projector & screen	2
3.	Printer	1
4.	Scanner	1
5.	Internet connection with Wi-Fi facility	1
6.	Whiteboard with accessories	2
7.	Sample/laboratory dyeing machine (infrared or glycerine bath, 8–12 pot)	1
8.	Laboratory padding mangle (padder) with pressure adjustment	1
9.	Sample dryer/relax dryer cum curing machine	1
10.	Flat-bed screen printing table with squeegee set (Lab Version)	1 set
11.	Screen frames, screens and squeegees (assorted mesh)	5 sets
12.	Print paste mixer/stirrer with homogenizer	1
13.	Colour kitchen arrangement with dissolving vessels, stirrers and dosing tanks	1 set
14.	Analytical balance (0.001 g accuracy)	1
15.	Electronic balance (0.01 g accuracy)	2
16.	Platform/bulk weighing balance	1
17.	Digital pH meter with buffer solutions	2
18.	Spectrophotometer with colour matching software and computer	1 set

19.	Light box/colour matching cabinet (D65, TL84, CWF, UV, incandescent A)	1
20.	Crock meter (rubbing fastness tester)	1
21.	Wash fastness tester (launder-o-meter/rota wash)	1
22.	Perspiration tester with oven	1
23.	Grey scale set for colour change and staining	2
24.	GSM cutter with GSM balance	2
25.	Shrinkage and spirality template with steel scale	2
26.	Hydrometer/Baume meter set	2
27.	Viscometer or flow cup for print paste	1
28.	Digital thermometer and temperature probe	5
29.	Moisture meter	2
30.	Laboratory glassware set (beaker, measuring cylinder, pipette, burette, funnel, flask)	5 sets
31.	Titration and laboratory chemical testing set	2
32.	Strength tester (tensile/tearing/bursting) where available	1
33.	Pilling and abrasion tester (where available)	1
34.	Fabric inspection machine with lighting arrangement	1
35.	Grey, pre-treated, dyed, printed and finished fabric samples (knit and woven)	As required
36.	Dyes – reactive, disperse, direct, vat, sulphur, acid and pigment	As required
37.	Chemicals and auxiliaries for pre-treatment, dyeing, printing and finishing	As required
38.	Chemical storage rack/cabinet with segregation arrangement	2
39.	Eye wash station and emergency safety shower	1 set
40.	Chemical spill kit with absorbent and neutralizing material	2
41.	Exhaust/ventilation arrangement for the laboratory and colour kitchen	1 set
42.	Soft water supply/water softening arrangement	1
43.	Waste segregation bins (colour coded)	4
44.	Tool board with hand tool set (spanner, screwdriver, plier, scissors)	5
45.	Fire extinguisher (appropriate types for chemical and electrical fire)	4
46.	First aid box with eye wash bottle	2
47.	PPE set (chemical goggles, chemical gloves, heat-resistant gloves, nose mask, apron, rubber boots, cap)	25 sets

The following conditions must be fulfilled –

1. The institute shall not use the same facilities for any other projects/organizations offering a similar course.
2. The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facilities required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirements of the Competency Standard (CS) provided by SICIP.

For the operation of the training course on **Production Process in Wet Processing**, the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training. Because full-scale wet processing machinery is capital-intensive, a training institute that does not possess production-scale machines must establish a formal linkage with a dyeing, printing and finishing unit so that the practical sessions of the occupation-specific units can be conducted in an actual workplace under supervision.

The institute will also provide all other hand tools, laboratory glassware and measuring instruments as per the CS for 25 trainees. The institute will also arrange adequate seating arrangements and classroom setup for the 25 trainees.